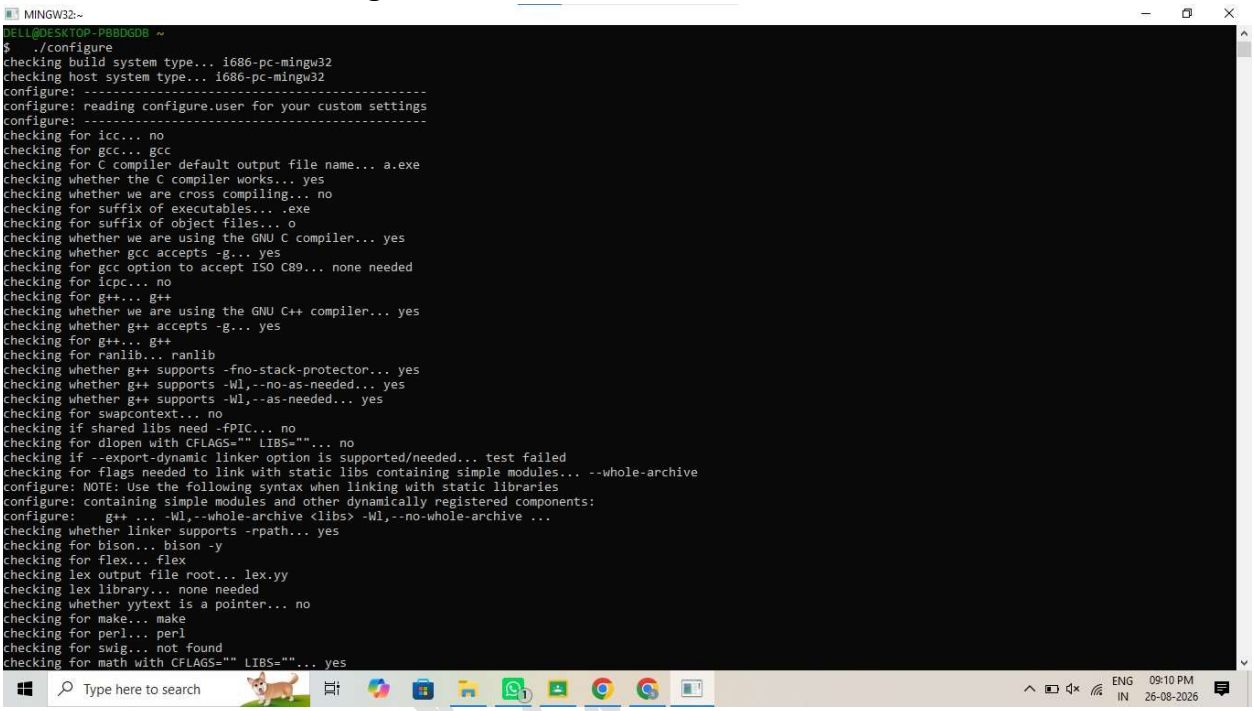
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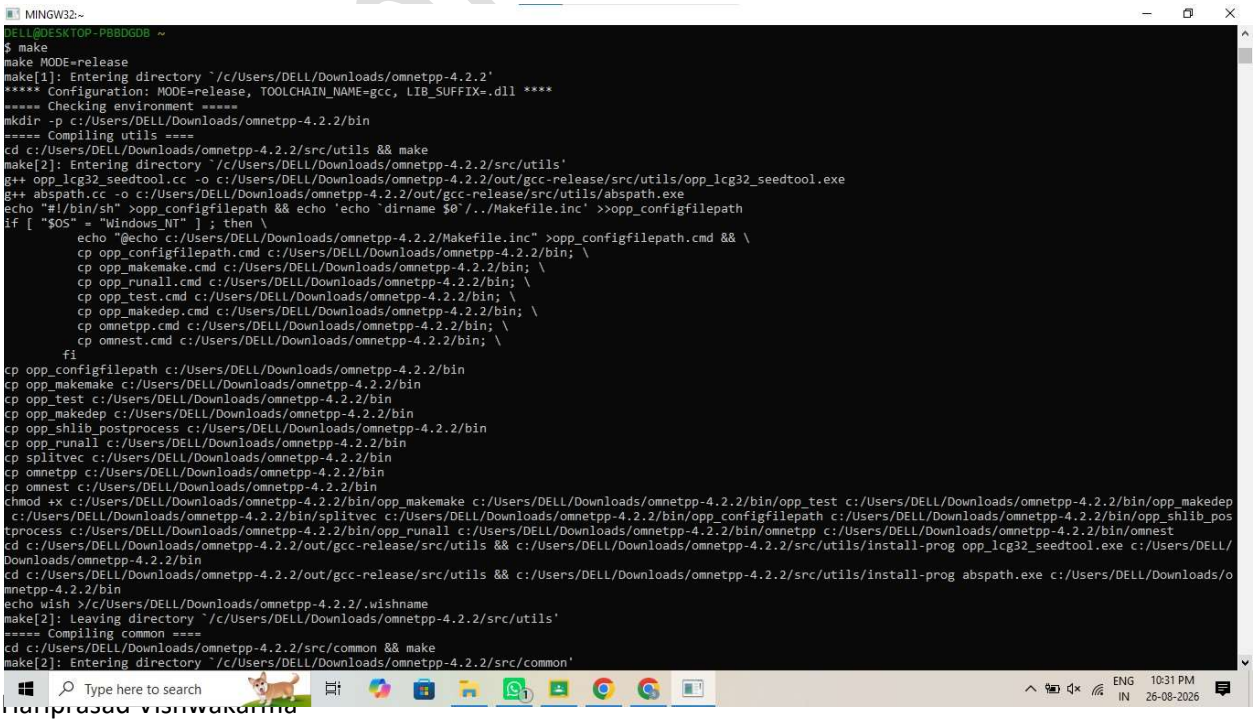
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Aim: Create a simple Adhoc network.

Enter command : ./configure



Enter command : make



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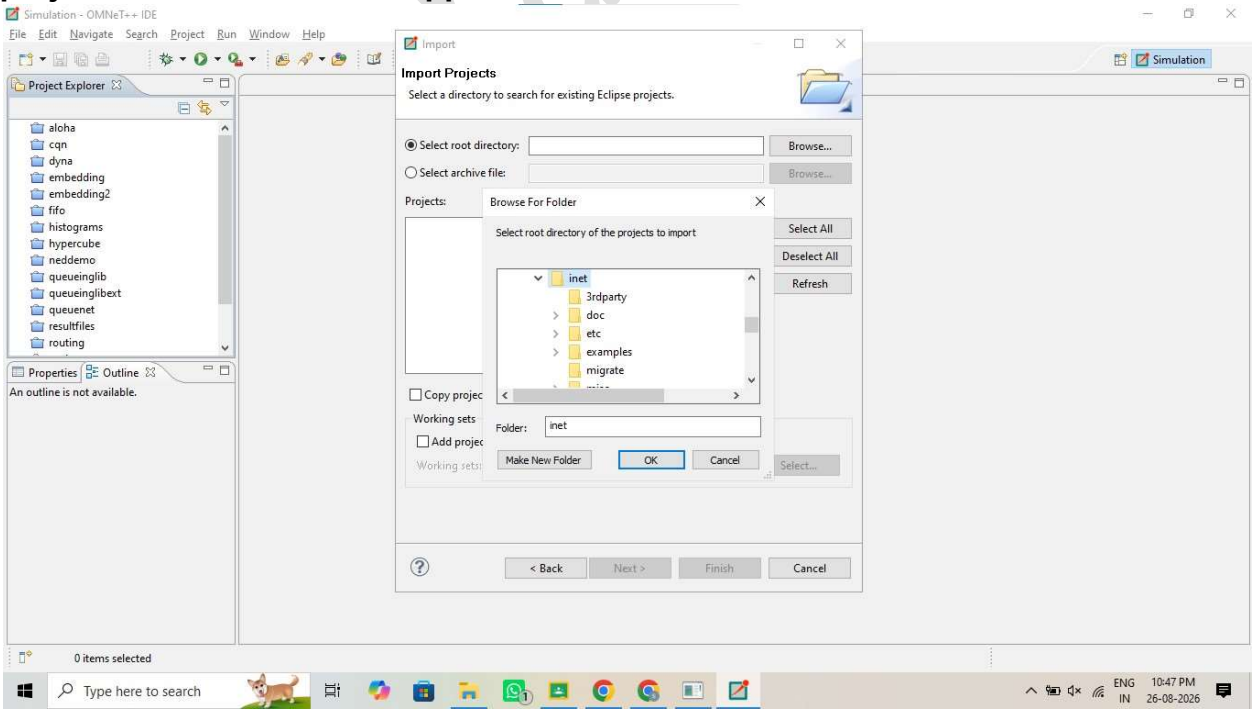
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Start Omnet++



import the project via File -> Import -> Existing Projects to the Workspace. A project named inet should appear.

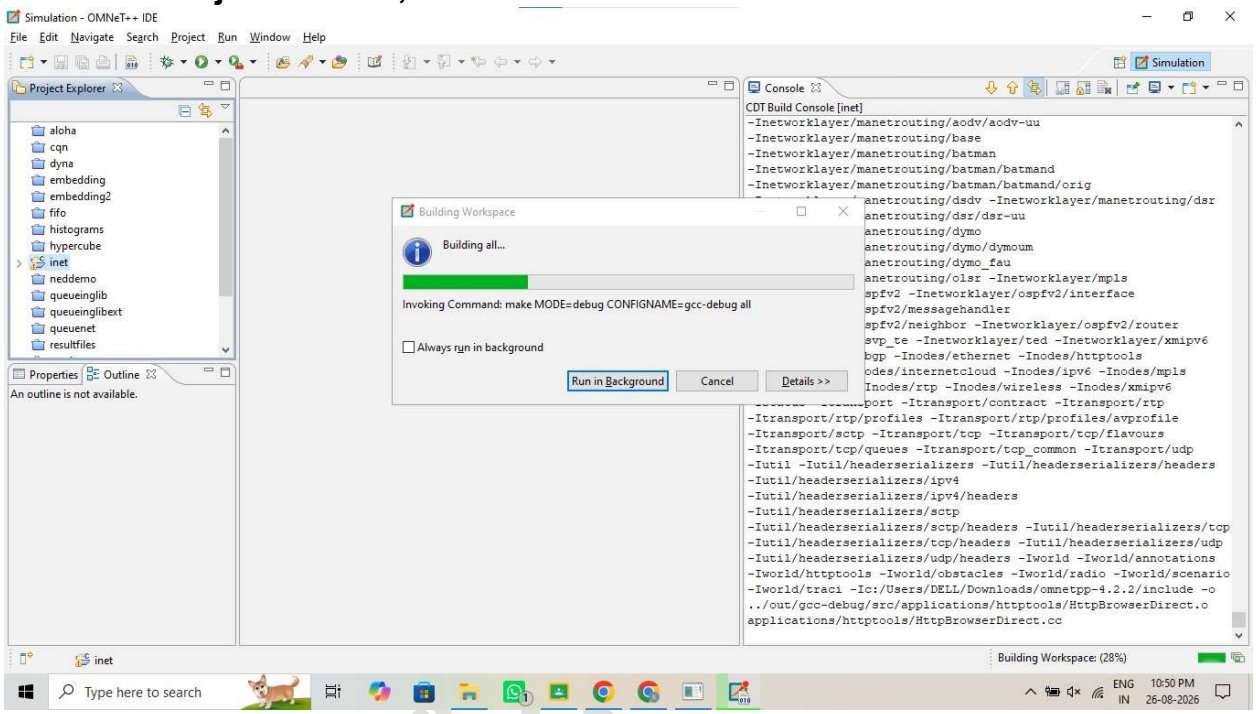


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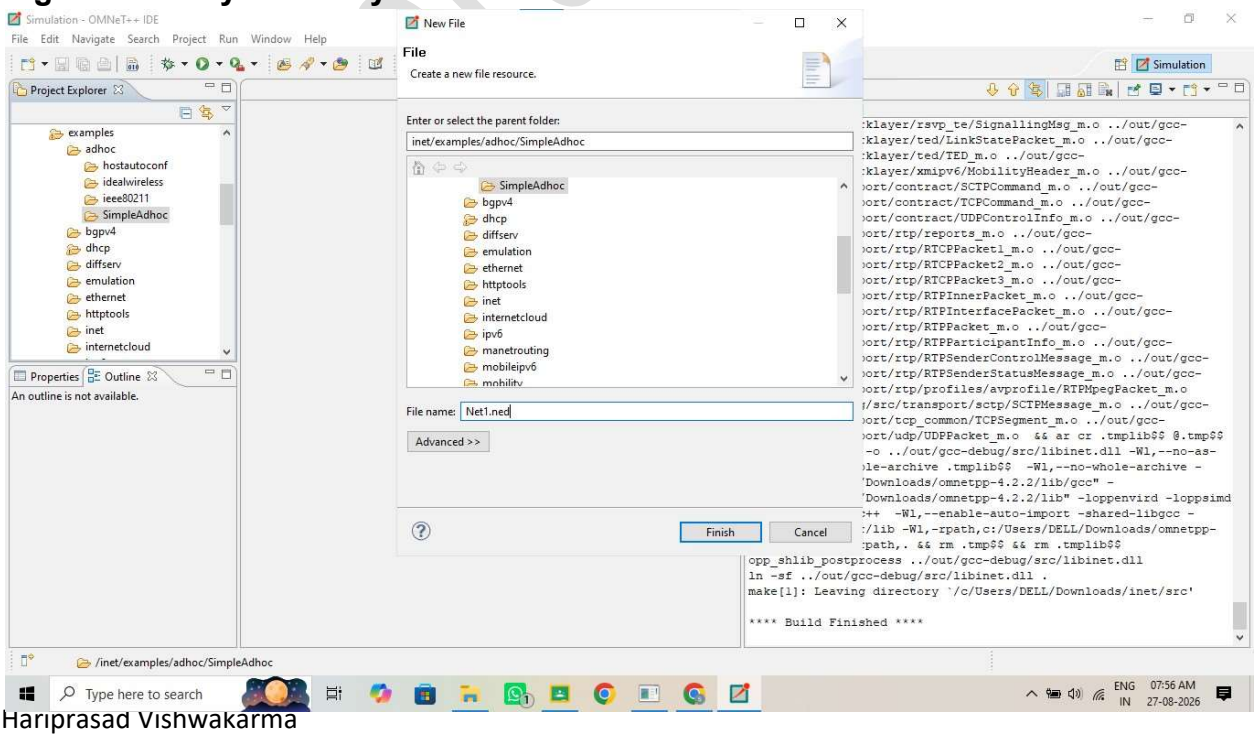
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Build with Project -> Build, or hit ctrl+b



Right click on your newly created folder and select NED file. Give name as Net1.



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```
[General]
network = Net1

*.numOfHosts = 5

# debug
#debug-on-errors = true
tkenv-plugin-path = ../../../etc/plugins

# constraint area
*.constraintAreaMinX = 0m
*.constraintAreaMinY = 0m
*.constraintAreaMinZ = 0m
*.constraintAreaMaxX = 600m
*.constraintAreaMaxY = 400m
*.constraintAreaMaxZ = 0m

*.debug = true
*.coreDebug = false

*.host*.**.channelNumber = 0

# channel physical parameters
*.channelControl.carrierFrequency = 2.4GHz
*.channelControl.pMax = 2.0mW
*.channelControl.sat = -110dBm
*.channelControl.alpha = 2
```

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```
# mobility
*.host*.mobilityType = "MassMobility"
*.host*.mobility.initFromDisplayString = false
*.host*.mobility.changeInterval = truncnormal(2s, 0.5s)
*.host*.mobility.changeAngleBy = normal(0deg, 30deg)
*.host*.mobility.speed = truncnormal(20mps, 8mps)
*.host*.mobility.updateInterval = 100ms

# ping app (host[0] pinged by others)
*.host[0].numPingApps = 0
*.host[*].numPingApps = 2
*.host[*].pingApp[*].destAddr = "host[0]"
*.host[0].pingApp[0].startTime = uniform(1s, 5s)
*.host[0].pingApp[1].startTime = 5s + uniform(1s, 5s)
*.host[*].pingApp[*].printPing = true

# nic settings
*.wlan[*].bitrate = 2Mbps




*.wlan[*].mgmt.frameCapacity = 10
*.wlan[*].mac.address = "auto"
*.wlan[*].mac.maxQueueSize = 14
*.wlan[*].mac.rtsThresholdBytes = 3000B
*.wlan[*].mac.retryLimit = 7
*.wlan[*].mac.cwMinData = 7

*.wlan[*].radio.transmitterPower = 2mW
```

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```
*.wlan[*].radio.thermalNoise = -110dBm
*.wlan[*].radio.sensitivity = -85dBm
*.wlan[*].radio.pathLossAlpha = 2
*.wlan[*].radio.snirThreshold = 4dB

[Config Ping1]
description = "host1 pinging host0"

[Config Ping2]
# __interactive__
description = "n hosts"

# leave numHosts undefined here

*.mobility.constraintAreaMinZ = 0m
*.mobility.constraintAreaMaxZ = 0m
*.mobility.constraintAreaMinX = 0m
*.mobility.constraintAreaMinY = 0m
*.mobility.constraintAreaMaxX = 600m
*.mobility.constraintAreaMaxY = 400m

*.debug = false
*.coreDebug = false
*.channelNumber = 0

# channel physical parameters
*.channelControl.carrierFrequency = 2.4GHz
*.channelControl.pMax = 20.0mW
```

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```
*.channelControl.sat = -110dBm
*.channelControl.alpha = 2

# mobility
**.host[*].mobilityType = "MassMobility"
**.host[*].mobility.changeInterval = truncnormal(2s, 0.5s)
**.host[*].mobility.changeAngleBy = normal(0deg, 30deg)
**.host[*].mobility.speed = truncnormal(20mps, 8mps)
**.host[*].mobility.updateInterval = 100ms

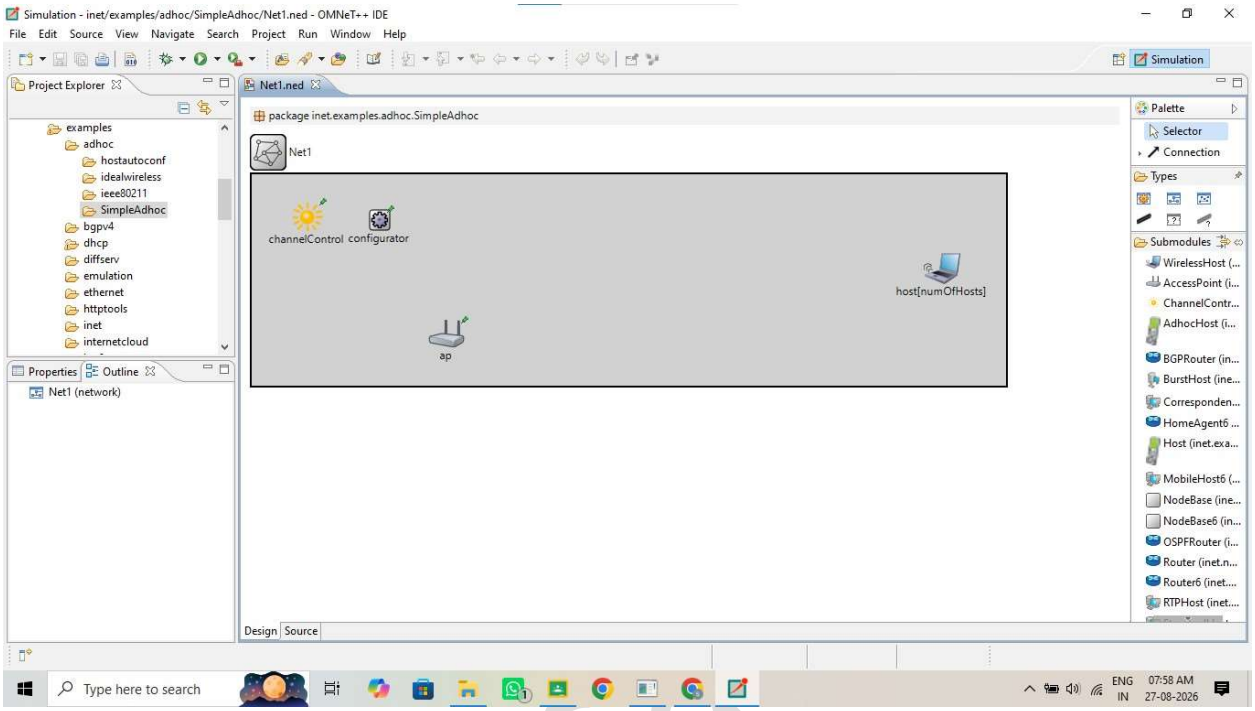
# nic settings
**.bitrate = 2Mbps
**.mac.address = "auto"
**.mac.maxQueueSize = 14
**.mac.rtsThresholdBytes = 3000B
**.wlan[*].mac.retryLimit = 7
**.wlan[*].mac.cwMinData = 7
**.wlan[*].mac.cwMinMulticast = 31
**.radio.transmitterPower = 20.0mW
**.radio.carrierFrequency = 2.4GHz
**.radio.thermalNoise = -110dBm
**.radio.sensitivity = -85dBm
**.radio.pathLossAlpha = 2
**.radio.snirThreshold = 4dB
# relay unit configuration
**.relayUnitType = "MACRelayUnitNP"
**.relayUnit.addressTableSize = 100
**.relayUnit.agingTime = 120s
**.relayUnit.bufferSize = 1MiB
**.relayUnit.highWatermark = 512KiB
**.relayUnit.pauseUnits = 300
# pause for 300*512 bit (19200 byte) time
**.relayUnit.addressTableFile = ""
**.relayUnit.numCPUs = 2
**.relayUnit.processingTime = 2us
```

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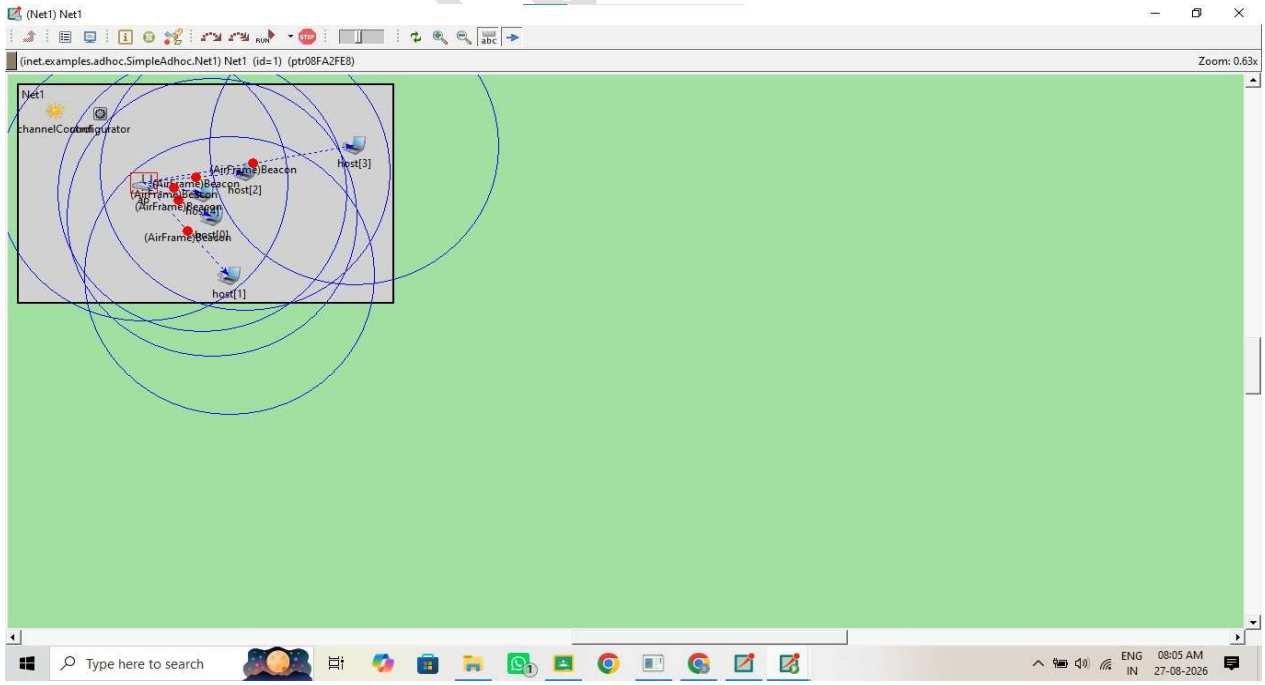
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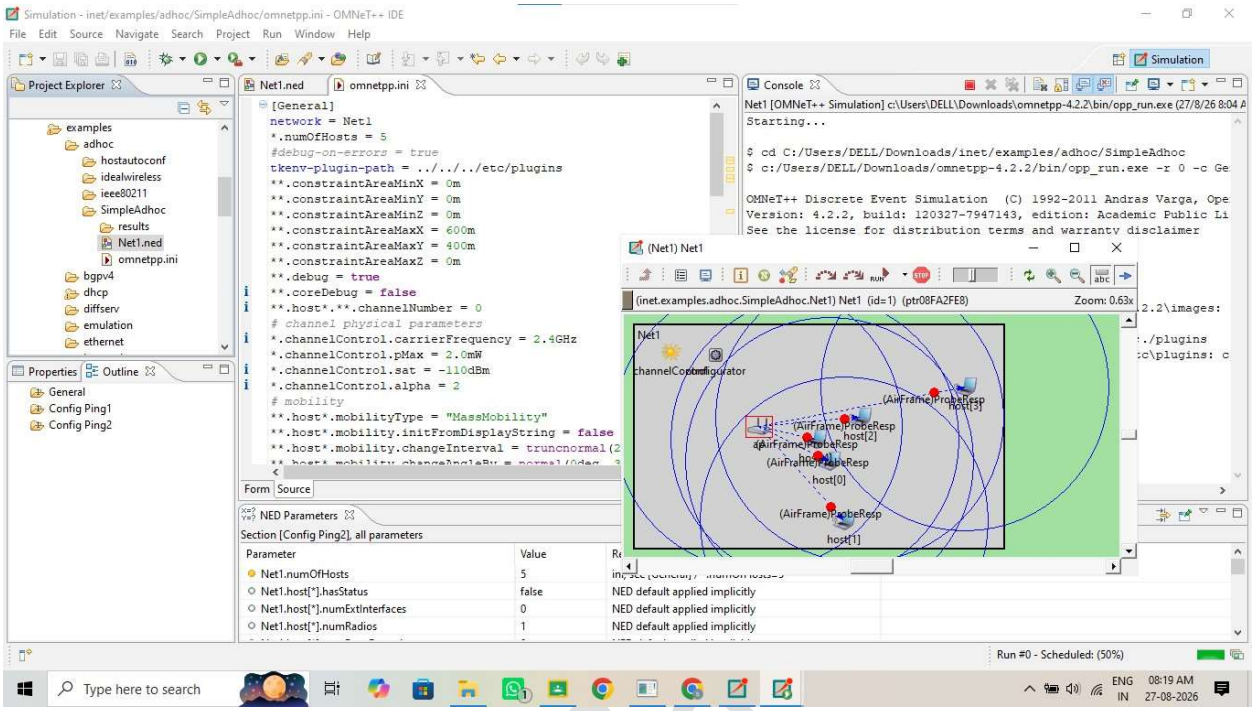
OUTPUT



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